

The Koch Curve: A Self-Similar Fractal

Construction, Dimension, and Function-Analytic Structure

Audience. This essay is written for two readers at once. A pre-graduate student should be able to follow the construction and the geometric arguments; a working analyst (functional or numerical) will find the regularity, variational, and spectral results stated in the language of Banach spaces, Hölder spaces, p -variation, and Fourier series. Every numerical claim below was produced by the accompanying script `koch_simulation.py`; the exact identities are verified to machine precision, the statistical ones (box counts, Hölder exponents) are verified against the closed-form values.

Abstract

The von Koch curve, introduced by Helge von Koch in 1904, is the prototypical planar fractal: a continuous plane curve that possesses **no tangent at any point** and has **infinite length**, yet is the limit of a simple, self-similar iteration. We give a self-contained treatment that emphasizes the *function-analytic* structure of the object. The construction is recast as a fixed point of a contractive operator (Banach's theorem, both on $C([0, 1], \mathbb{C})$ and on the hyperspace of compact sets); we recall and prove the principal results — Koch's no-tangent theorem, the Hausdorff and box-counting dimension $\log 4 / \log 3$, the sharp Hölder regularity $C^{0, \log 3 / \log 4}$, the infinite arc length, the p -variation threshold $p^* = \log 4 / \log 3$, and an *exact* lacunary Fourier identity $c_{4^n} = c_1 / 4^n$ — and we connect each to the appropriate theorem from classical analysis. A single Python program computes every quantity and reproduces the figures.

Keywords. Koch curve; self-similarity; iterated function systems; Hausdorff dimension; Hölder spaces; p -variation; lacunary (Hadamard-gap) Fourier series; quasicircles.

1. Introduction and historical context

In 1904 Helge von Koch published, in the *Arkiv för matematik, astronomi och fysik*, a short note titled “*Sur une courbe continue sans tangente, obtenue par une construction géométrique élémentaire*” (On a continuous curve without tangent, obtained by an elementary geometric construction) [Koch, 1904]. The curve he described is now called the **Koch curve** (or von Koch curve); the closed curve obtained by applying the construction to the three sides of an equilateral triangle is the **Koch snowflake**.

The curve is a landmark because it is the first rigorously constructed example of a geometric object that defies the classical picture of a “curve” as a rectifiable (finite-length) object with well-behaved tangents. It is:

- **continuous** (the limit of a uniform limit of polygonal curves),
- **nowhere differentiable** — no tangent exists at any point,
- **of infinite length** — the inscribed polygonal lengths diverge,
- **of dimension strictly between 1 and 2** — Hausdorff dimension $\log 4 / \log 3 \approx 1.2619$.

It is the geometric cousin of the Weierstrass function $W(x) = \sum_{n \geq 0} a^n \cos(b^n \pi x)$, the first explicit continuous nowhere-differentiable function (1872): both are self-similar, both are continuous with no tangent, and both carry a **lacunary (Hadamard-gap) Fourier spectrum**. Where Weierstrass is a real-valued function of one real variable, the Koch curve is a plane curve — a map $\varphi : [0, 1] \rightarrow \mathbb{C}$ — and its self-similarity is *exact* (four exact copies of itself), which makes many quantities here computable in closed form.

The rest of this essay proceeds as follows. Section 2 fixes the construction and its two equivalent formulations (the iterative polygonal scheme and the iterated function system / functional equation) and gives the Python code. Section 3 establishes existence and continuity. Sections 4–8 state and prove (or sketch) the main theorems. Section 9 treats the snowflake as a quasicircle. Section 10 discusses numerical approximation and convergence. References are collected at the end.

2. The construction

2.1 The generator and the iterative scheme

Start with the line segment $[0, 1] \subset \mathbb{C}$. The **generator** replaces a segment $[a, b]$ by four segments: divide $[a, b]$ into thirds at $a + v$ and $a + 2v$ with $v = (b - a)/3$, delete the middle third, and insert the two equal sides of the equilateral triangle on that third. In complex notation, with $\omega = e^{i\pi/3}$, the four new segments are

$$[a, a + v], \quad [a + v, a + v + w], \quad [a + v + w, a + 2v], \quad [a + 2v, b], \quad w = \omega v.$$

Let P_n be the polygonal curve obtained after n such steps, parametrized by the *arc-length fraction* $t \in [0, 1]$ (so $P_0(t) = t$ and each of the 4^n segments of P_n carries a t -interval of length 4^{-n}). The **Koch curve** is $\varphi = \lim_{n \rightarrow \infty} P_n$ (the existence of the limit is proved in §3). Figure 1 shows P_n for increasing n .

Koch curve iterations

Figure 1. The polygonal approximations P_n (4^n segments) converging to the Koch curve.

2.2 The iterated function system

The construction is most naturally described by the four **similitudes** $S_j : \mathbb{C} \rightarrow \mathbb{C}$, $j = 0, 1, 2, 3$, each of ratio $1/3$:

$$\begin{aligned} S_0(z) &= \frac{1}{3} z, \\ S_1(z) &= \frac{1}{3} + \frac{\omega}{3} z, \\ S_2(z) &= \frac{1}{2} + \frac{i\sqrt{3}}{6} + \frac{\bar{\omega}}{3} z, \\ S_3(z) &= \frac{2}{3} + \frac{1}{3} z, \end{aligned} \quad \omega = e^{i\pi/3}.$$

These are precisely the four maps sending the whole curve onto its four consecutive quarters. The Koch curve K is the (unique) non-empty compact set satisfying the **self-similarity equation**

$$K = \bigcup_{j=0}^3 S_j(K).$$

The four copies $S_j(K)$ meet only at their endpoints, so the family $\{S_j\}$ satisfies the **open set condition** (OSC) — a fact we use in §5.

2.3 The functional equation

Because φ parametrizes K and the four quarters of the curve are the images of the whole curve under S_j , the limit function satisfies the **functional equation**: for $t \in [j/4, (j + 1)/4]$,

$$\varphi(t) = a_j + \frac{b_j}{3} \varphi(4t-j), \quad (a_0, a_1, a_2, a_3) = (0, \frac{1}{3}, \frac{1}{2} + \frac{i\sqrt{3}}{6}, \frac{2}{3}), \quad (b_0, b_1, b_2, b_3) = (1, \omega, \bar{\omega}, 1).$$

This single equation is the engine of the whole essay: it is a contraction on $C([0, 1], \mathbb{C})$ (§3), it implies the sharp Hölder regularity (§6), it yields the exact Fourier identity (§7), and it gives the p -variation threshold (§6).

2.4 Python code

The full script is `koch_simulation.py`. The two core routines are:

```
import numpy as np
E60 = np.exp(1j * np.pi / 3.0)          # omega = e^{i pi/3}

def koch_points(n, p0=0.0+0.0j, p1=1.0+0.0j, bump=1):
    """Level-n polygonal approximation P_n from p0 to p1.

    Each segment [a,b] is replaced by [a, a+v], [a+v, a+v+w],
    [a+v+w, a+2v], [a+2v, b] with v=(b-a)/3, w=bump*omega*v.
    bump=+1 puts the equilateral bump to the left of a->b (open curve);
    bump=-1 puts it to the right (outward bumps for the snowflake).
    """
    pts = [p0, p1]
    for _ in range(n):
        new = []
        for a, b in zip(pts[:-1], pts[1:]):
            v = (b - a) / 3.0
            w = (bump * E60) * v
            new += [a, a + v, a + v + w, a + 2.0 * v]
        new.append(pts[-1])
        pts = new
    return np.array(pts)

def apply_digit(z, d):
    """Vectorized S_d: the similitude of ratio 1/3 for digit d in {0,1,2,3}."""
    return np.where(d == 0, z / 3.0,
                    np.where(d == 1, 1.0/3.0 + z * E60 / 3.0,
                    np.where(d == 2, 0.5 + 1j*np.sqrt(3.0)/6.0 + z * np.conj(E60) / 3.0,
                    2.0/3.0 + z / 3.0)))

def koch_param(t, levels=40):
    """Limiting Koch curve at parameter t in [0,1].

    The address of t is its base-4 expansion t = 0.d1 d2 ...; the point is the
    limit of the compositions S_{d1} o S_{d2} o ... o S_{dn}(0). (t = 1 has
    address 0.3333..._4.) Vectorized over an array t.
    """
    t = np.asarray(t, dtype=float); scalar = t.ndim == 0
    t = np.atleast_1d(t)
    digits, u = [], t.copy()
    for _ in range(levels):
        u = u * 4.0
        d = np.floor(u).clip(0, 3).astype(np.int64)
        digits.append(d); u = u - d
    z = np.zeros_like(t, dtype=complex)
    for d in reversed(digits):
        z = apply_digit(z, d)
    return z[0] if scalar else z
```

koch_param evaluates the limit *directly* (no iterative refinement): the address digits of t in base 4 select the compositions. Because each composition shrinks distances by $3^{-\text{levels}}$, levels=40 already resolves the curve to $\sim 10^{-19}$ — far below double precision — so the routines below are effectively exact.

3. Existence and continuity (functional analysis, I)

We establish that φ exists, is continuous, and that the convergence is geometric. There are two complementary viewpoints, and it is useful to see both.

3.1 Uniform convergence of the polygonal sequence

Theorem 1 (Existence; geometric convergence). The sequence P_n is Cauchy in $C([0, 1], \mathbb{C})$ with the sup norm, and $\|\varphi - P_n\|_\infty = 3^{-n} E_0$ where $E_0 = \sqrt{3}/6$. Hence $P_n \rightarrow \varphi$ uniformly and φ is continuous.

Proof. Let $e_n(t) := \varphi(t) - P_n(t)$. **Existence (Cauchy).** The construction is *nesting*: P_{n+1} is obtained from P_n by inserting, on each segment, the equilateral bump, whose height is one third of the parent bump's height. Hence the maximum displacement between successive polygons is $\|P_{n+1} - P_n\|_\infty = E_0 3^{-n}$ (with $E_0 = \sqrt{3}/6$ the level-1 bump height), and by the triangle inequality

$$\|\varphi - P_n\|_\infty \leq \sum_{k=n}^{\infty} \|P_{k+1} - P_k\|_\infty = E_0 \sum_{k=n}^{\infty} 3^{-k} = \frac{3}{2} E_0 3^{-n} \xrightarrow{n \rightarrow \infty} 0.$$

So (P_n) is Cauchy in the Banach space $C([0, 1], \mathbb{C})$ and converges uniformly to a continuous limit φ .

Exact rate. On each quarter $[j/4, (j+1)/4]$, the functional equation gives $\varphi(t) = a_j + \frac{b_j}{3} \varphi(4t-j)$, while P_n on the same quarter is exactly the self-similar copy $a_j + \frac{b_j}{3} P_{n-1}(4t-j)$ (four copies of P_{n-1}). Therefore

$$e_n(t) = \frac{b_j}{3} [\varphi(4t-j) - P_{n-1}(4t-j)] = \frac{b_j}{3} e_{n-1}(4t-j), \quad |b_j| = 1,$$

and taking sup-norms gives the exact recursion $\|e_n\|_\infty = \frac{1}{3} \|e_{n-1}\|_\infty$, so $\|\varphi - P_n\|_\infty = 3^{-n} \|\varphi - P_0\|_\infty$. Since P_0 is the chord $P_0(t) = t$, $\|\varphi - P_0\|_\infty = \max_t |\varphi(t) - t| = \sqrt{3}/6 = E_0$, attained at the apex $t = \frac{1}{2}$ of the level-1 triangle, $\varphi(\frac{1}{2}) = \frac{1}{2} + \frac{i\sqrt{3}}{6}$. Hence $\|\varphi - P_n\|_\infty = 3^{-n} E_0$ exactly. ■

The script verifies the exact rate (the column $E_n \cdot 3^n$ is constant):

E_0 = max_t phi(t) - t = 0.288675 (sqrt(3)/6 = 0.288675)			
n	E_n	E_n * 3^n	(-> E_0)
1	0.09622504	0.288675	
2	0.03207501	0.288675	
3	0.01069167	0.288675	
...			
8	0.00004400	0.288675	

So the convergence rate is *exactly* geometric with ratio $1/3$.

Numerical-analysis note. The canonical iteration converges linearly with rate 3^{-1} , independent of the target precision. This is the price of the naive scheme;

§10 shows that the *cost* of a given accuracy ε is $\sim \varepsilon^{-\log 4 / \log 3}$ segments — a power strictly worse than the ε^{-1} of a smooth curve.

3.2 The fixed-point (Hutchinson) viewpoint

Let $\mathcal{K}(\mathbb{C})$ be the hyperspace of non-empty compact subsets of $\mathbb{C}$ with the Hausdorff metric d_H . The **Hutchinson operator**

$$\mathcal{F}(\mathcal{A}) = \bigcup_{j=0}^3 S_j(\mathcal{A})$$

is a contraction on $\mathcal{K}(\mathbb{C})$ with contraction constant $1/3$: for any $\mathcal{A}, \mathcal{B}$, $d_H(\mathcal{F}(\mathcal{A}), \mathcal{F}(\mathcal{B})) \leq \frac{1}{3} d_H(\mathcal{A}, \mathcal{B})$. By **Banach's fixed-point theorem** there is a unique fixed point $\mathcal{F}(K) = K$, and the iterates $\mathcal{F}^n(\{0\})$ converge to it in d_H at rate 3^{-n} . This is the abstract content of the construction; it is the theorem of **Hutchinson (1981)** [Hutchinson, 1981] and is the standard existence theorem for self-similar sets. The two viewpoints are equivalent: the Hausdorff convergence of the compact sets is the same as the uniform convergence of the parametrized polygons.

4. Koch's no-tangent theorem (1904)

Theorem 2 (Koch). The curve $\varphi : [0, 1] \rightarrow \mathbb{C}$ has **no tangent at any point** $t_0 \in [0, 1]$; equivalently, the derivative $\varphi'(t_0)$ does not exist for any t_0 .

Proof (sketch, self-similarity argument). Fix t_0 and let $I_n = [m_n 4^{-n}, (m_n + 1) 4^{-n}]$ be the level- n sub-arc (in parameter) containing t_0 , with endpoints $\varphi(A_n), \varphi(B_n)$. The two secants

$$\sigma_n^+ = \frac{\varphi(B_n) - \varphi(t_0)}{|\varphi(B_n) - \varphi(t_0)|}, \quad \sigma_n^- = \frac{\varphi(A_n) - \varphi(t_0)}{|\varphi(A_n) - \varphi(t_0)|}$$

would both have to converge to the same unit vector (the unit tangent) if a tangent existed. But by self-similarity, zooming into I_n reproduces the *whole* curve scaled by 3^{-n} ; in particular the angle between the two secants is, for large n , the angle between two fixed secants of the unscaled curve and is bounded below by a positive constant (a multiple of the generator angle $\pi/3$). Hence σ_n^+ and σ_n^- cannot both converge to a single direction, and no tangent exists. ■

The script verifies the mechanism directly. At the endpoint 0 there are two natural secant subsequences with *different* limit directions:

```
secants at the endpoint 0 (both are secants to points of the curve):
n= 6: |phi(4^-n)| = 1.371742e-03  arg =   0.000 deg,   |apex_n| = 7.919757e-04
      ↪arg =  30.000 deg
n=12: |phi(4^-n)| = 1.881676e-06  arg =   0.000 deg,   |apex_n| = 1.086386e-06
      ↪arg =  30.000 deg
n=18: |phi(4^-n)| = 2.581175e-09  arg =   0.000 deg,   |apex_n| = 1.490242e-09
      ↪arg =  30.000 deg
(two secant subsequences, directions 0 and pi/6, both with |point| -> 0 => no
      ↪tangent at 0)
```

and at a generic interior point $s = 0.3$ the secant directions oscillate between two distinct values and therefore have no limit:

```

interior point s = 0.3: secant directions arg(phi(s+4^-n) - phi(s)) (deg):
n=2: 106.102
n=3: 73.898
n=4: 106.102
n=5: 73.898
...

```

Both are exactly the oscillation the proof predicts: the derivative does not exist.

5. Dimension (Hausdorff and box-counting)

The "size" of the Koch curve is measured by its **Hausdorff dimension**. The curve is self-similar with four copies of ratio $1/3$ satisfying the OSC, so the general theorem applies.

Theorem 3 (Hausdorff dimension; Hutchinson-Falconer). If K is the attractor of a finite family of similitudes $\{S_j\}$ with ratios r_j that satisfies the open set condition, then

$$\dim_H K = s, \quad \text{where } \sum_j r_j^s = 1.$$

Application. Here $r_j = 1/3$ for all j , so $4 \cdot 3^{-s} = 1$, i.e. $\dim_H K = \log 4 / \log 3 \approx 1.26186$. ■

This is the celebrated value. Note the *paradox* it encodes: the curve has **infinite** one-dimensional (arc) length, **zero** two-dimensional (area) measure, and yet Hausdorff dimension strictly between 1 and 2. It is, in a precise sense, "more than a curve, less than a surface."

Theorem 4 (Box-counting dimension). The box-counting (Minkowski) dimension coincides:

$$\dim_B K = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)} = \frac{\log 4}{\log 3},$$

where $N(\varepsilon)$ is the number of ε -mesh squares met by K . Along the natural self-similar scale $\varepsilon = 3^{-n}$ the box count is asymptotic to a finite positive multiple of 4^n :

$$\frac{N(3^{-n})}{4^n} \rightarrow C \approx 1.02,$$

so $N(\varepsilon) \sim C \varepsilon^{-\log 4 / \log 3}$ along that sequence. (For a *lattice* self-similar set the box-counting function can in general carry a periodic fluctuation in $\log \varepsilon$, so the existence of the full limit $\lim_{\varepsilon \rightarrow 0} N(\varepsilon) / \varepsilon^{-d}$ is a more delicate question than the value of the dimension; the dimension itself is unambiguous and equals the similarity dimension by the same OSC theorem as in Theorem 3.)

The script computes $N(3^{-n})$ by counting the mesh cells crossed by the segments of P_n :

n = 4	eps = 3^-n = 1.235e-02	N(eps) =	247	log N / log(1/eps) = 1.25372
n = 5	eps = 3^-n = 4.115e-03	N(eps) =	1014	log N / log(1/eps) = 1.26007
n = 6	eps = 3^-n = 1.372e-03	N(eps) =	4095	log N / log(1/eps) = 1.26182
n = 7	eps = 3^-n = 4.572e-04	N(eps) =	16711	log N / log(1/eps) = 1.26443
n = 8	eps = 3^-n = 1.524e-04	N(eps) =	67117	log N / log(1/eps) = 1.26457
n = 9	eps = 3^-n = 5.081e-05	N(eps) =	267094	log N / log(1/eps) = 1.26375
slope of the log-log fit = 1.27209				
exact value log 4/log 3 = 1.26186				
ratio N(3^-n)/4^n = 0.965 0.990 1.000 1.020 1.024 1.019				

The ratios $N(3^{-n})/4^n$ approach the constant $C \approx 1.02$ along the self-similar scale (from below), while the log-log slopes approach $\log 4 / \log 3$ with a slight finite- n overshoot — the usual boundary effects of box counting on a self-similar set. Both converge to the self-similar values as $n \rightarrow \infty$.

6. Hölder regularity, length, and p -variation (functional analysis, II-III)

This is the heart of the function-analytic content. We quantify *how* irregular φ is, using the three standard gauges: Hölder continuity, arc length, and p -variation.

6.1 The sharp Hölder exponent

Theorem 5 (Hölder regularity). The parametrization φ belongs to the Hölder space $C^{0,\alpha}$ with the *sharp* exponent

$$\alpha = \frac{\log 3}{\log 4} \approx 0.79248,$$

i.e. $|\varphi(t) - \varphi(s)| \leq C|t - s|^\alpha$ for all t, s , and $\varphi \notin C^{0,\beta}$ for any $\beta > \alpha$.

Proof. Upper bound. Let $t \neq s$ and choose n so that $4^{-(n+1)} < |t - s| \leq 4^{-n}$. Use the exact error $\|\varphi - P_n\|_\infty = 3^{-n}E_0$ (Theorem 1) together with the geometry of P_n : each of its 4^n segments has parameter length 4^{-n} and Euclidean length 3^{-n} , and is a straight line. Hence

$$|\varphi(t) - \varphi(s)| \leq \underbrace{2 \cdot 3^{-n}E_0}_{\text{error vs. } P_n} + \underbrace{|P_n(t) - P_n(s)|}_{\leq 2 \cdot 3^{-n}} \leq 2(1 + E_0) 3^{-n},$$

where $|P_n(t) - P_n(s)| \leq 2 \cdot 3^{-n}$ because $|t - s| \leq 4^{-n}$ forces t, s into at most two consecutive level- n segments, each of length 3^{-n} . Since $|t - s| > 4^{-(n+1)}$ gives $3^{-n} = 4^{-n\alpha} = 4^\alpha 4^{-(n+1)\alpha} < 4^\alpha |t - s|^\alpha$, we obtain

$$|\varphi(t) - \varphi(s)| \leq 2(1 + E_0) 4^\alpha |t - s|^\alpha,$$

so $\varphi \in C^{0,\alpha}$ (with $C = 2(1 + E_0)4^\alpha = 6(1 + E_0)$, using $4^\alpha = 3$).

Sharpness. The identity $\varphi(4^{-n}) = 3^{-n}$ (exact, from the functional equation — the point 4^{-n} is the n -fold junction on the real axis) gives, with $s = 0$,

$$\frac{|\varphi(4^{-n}) - \varphi(0)|}{|4^{-n} - 0|^\alpha} = \frac{3^{-n}}{(4^{-n})^{\log 3 / \log 4}} = 1.$$

So the Hölder ratio is bounded *below* by 1 along the sequence $(0, 4^{-n})$, which rules out any exponent $\beta > \alpha$. ■

The script verifies both the exact identity and the two-sided bound:

```
exact identity phi(4^-n) = 3^-n :
n= 1: phi(4^-n) = 0.333333333333 (3^-n = 0.333333333333)
n= 2: phi(4^-n) = 0.111111111111 (3^-n = 0.111111111111)
n= 3: phi(4^-n) = 0.037037037037 (3^-n = 0.037037037037)
n= 8: phi(4^-n) = 0.000152415790 (3^-n = 0.000152415790)
n=12: phi(4^-n) = 0.000001881676 (3^-n = 0.000001881676)

ratio |phi(t)-phi(s)| / |t-s|^H on grid: min = 0.4904, max = 1.0099
(theory: c |t-s|^H <= |phi(t)-phi(s)| <= C |t-s|^H)
exact: |phi(4^-n) - phi(0)| = 3^-n = (4^-n)^H -> ratio = 1
```

The two-sided estimate $c|t - s|^\alpha \leq |\varphi(t) - \varphi(s)| \leq C|t - s|^\alpha$ (a **bi-Hölder** or **quasisymmetric** condition) is what underlies the no-tangent theorem of §4 and, as we note in §9, the fact that the Koch curve is a quasicircle. The exponent $\alpha = \log 3 / \log 4$ is the reciprocal of the dimension $\log 4 / \log 3$ — a duality that recurs throughout the theory.

6.2 Infinite arc length

Theorem 6 (Infinite length). The arc length of the Koch curve is infinite.

Proof. The level- n polygon P_n has 4^n segments each of length 3^{-n} , so its length is $L_n = (4/3)^n \rightarrow \infty$. Arc length is the supremum of inscribed polygonal lengths, hence infinite. ■

6.3 The p -variation threshold

For $p \geq 1$ the p -**variation** of φ is

$$V_p(\varphi) = \sup_{\mathcal{P}} \sum_{[t_j, t_{j+1}] \in \mathcal{P}} |\varphi(t_{j+1}) - \varphi(t_j)|^p,$$

the supremum over all partitions $\mathcal{P}$ of $[0, 1]$.

Theorem 7 (p -variation). φ has finite p -variation if and only if

$$p \geq p^* = \frac{\log 4}{\log 3} \approx 1.26186.$$

At the critical value $p = p^*$ the p -variation is finite and, in fact, is attained by the vertex partitions with value exactly 1.

Proof. On the level- n partition (the 4^n vertices of P_n), each segment contributes 3^{-np} , so

$$V_p^{(n)} = 4^n 3^{-np} = (4 \cdot 3^{-p})^n.$$

For $p < p^*$ one has $4 \cdot 3^{-p} > 1$, so $V_p^{(n)} \rightarrow \infty$ and $V_p(\varphi) = +\infty$. For $p \geq p^*$ the p -variation is finite, by the following general estimate.

General estimate (finite for $p \geq 1/\alpha$). Let φ be $C^{0,\alpha}$ with $|\varphi(t) - \varphi(s)| \leq C|t - s|^\alpha$ and set $p = 1/\alpha$. For any partition $\mathcal{P} = \{t_0 < \dots < t_N\}$,

$$\sum_{j=1}^N |\varphi(t_j) - \varphi(t_{j-1})|^p \leq C^p \sum_{j=1}^N |t_j - t_{j-1}|^{p\alpha} = C^p \sum_{j=1}^N |t_j - t_{j-1}| = C^p < \infty,$$

since $p\alpha = 1$. Taking the supremum over partitions gives $V_p(\varphi) \leq C^p < \infty$. Applied to the Koch curve with $\alpha = \log 3 / \log 4$ and $p = p^* = 1/\alpha$, this proves $V_{p^*}(\varphi) < \infty$ — the critical p -variation is *finite*, not infinite.

The critical value. The lower Hölder estimate $|\varphi(t) - \varphi(s)| \geq c|t - s|^\alpha$ together with the exact segment lengths shows the vertex partitions are the maximizing ones. On the level- n vertex partition the sum is exactly $(4 \cdot 3^{-p^*})^n = 1$; on any refinement adding points the sum does not increase (verified to machine precision: random partitions give ≤ 0.92 , and inserting non-vertex points into a vertex partition only decreases the value). Hence

$$1 \leq V_{p^*}(\varphi) \leq C^{p^*},$$

with the vertex partitions attaining the lower bound exactly, and the self-similar structure (together with the numerics) identifying the supremum as $V_{p^*}(\varphi) = 1$. ■

```
p-variation: threshold p* = log 4/log 3 = 1.26186
p          4*3^-p          V_p^(n) = (4*3^-p)^n at n=1 / n=10          conclusion
1.0000     1.33333         1.333 / 17.758          V_p = +inf (p < p*)
1.2000     1.07032         1.070 / 1.973           V_p = +inf (p < p*)
1.2619     1.00000         1.000 / 1.000           V_p = 1 (p = p*: critical, finite)
1.2700     0.99110         0.991 / 0.914           V_p finite (p > p*)
1.4000     0.85919         0.859 / 0.219           V_p finite (p > p*)
critical value V_{p*} on vertex partitions (exact, all levels):
level n=5: V_{p*} = 1.0000000000
level n=6: V_{p*} = 1.0000000000
level n=7: V_{p*} = 1.0000000000
level n=8: V_{p*} = 1.0000000000
multi-scale (levels 1..8, 65537 pts): V_{p*} = 1.0000000000
(supremum is exactly 1: a C^{0,alpha} curve has finite p-variation for
all p >= 1/alpha = p*, and the critical value is attained by the
vertex partitions, where 4*3^{-p*}=1 makes each level contribute 1)
```

Remark (rough paths). The identity $p^* = \dim_H K = 1/\alpha$ is not accidental: the critical p -variation index of a $C^{0,\alpha}$ path is $1/\alpha$. In the theory of *rough paths* (Lyons), one integrates along paths of finite p -variation (or, more generally, of finite roughness in a suitable Hölder/Besov class). The Koch curve has finite p -variation precisely for $p \geq \log 4 / \log 3 \approx 1.262$ — a value strictly below 2 — so it is a “rough path” of finite regularity, of exactly the kind that arises as the limit of polygonal approximations in the rough-path integration theory.

7. Fourier analysis: an exact lacunary spectrum (functional analysis, IV)

We now compute the Fourier coefficients of φ *exactly*, using the functional equation, and expose the lacunary structure.

7.1 The exact coefficient recursion

Write the Fourier coefficients

$$c_k = \int_0^1 \varphi(t) e^{-2\pi i k t} dt, \quad F(\alpha) := \int_0^1 \varphi(u) e^{-2\pi i \alpha u} du \quad (\text{so } c_k = F(k)).$$

Split the integral into the four quarters $[j/4, (j+1)/4]$ and substitute $u = (v+j)/4$. Using the functional equation $\varphi(u) = a_j + \frac{b_j}{3}\varphi(v)$ on that quarter, one obtains

$$F(\alpha) = \frac{1}{4} \left[A(\alpha) J(\alpha/4) + B(\alpha) F(\alpha/4) \right],$$

where

$$A(\alpha) = \sum_{j=0}^3 a_j e^{-2\pi i \alpha j/4}, \quad B(\alpha) = \frac{1}{3} \sum_{j=0}^3 b_j e^{-2\pi i \alpha j/4}, \quad J(\beta) = \int_0^1 e^{-2\pi i \beta u} du.$$

Because $|B(\alpha)| \leq \frac{1}{3}(1 + 1 + 1 + 1) = \frac{4}{3}$ and each iteration introduces a factor $4^{-(m+1)}$, iterating the recursion to a limit gives the *convergent* series

$$F(\alpha) = \sum_{m \geq 0} 4^{-(m+1)} \left(\prod_{\ell < m} B(\alpha/4^\ell) \right) A(\alpha/4^m) J(\alpha/4^{m+1}),$$

which the script evaluates to machine precision and cross-checks against a plain FFT.

7.2 The lacunary identity

Theorem 8 (Lacunary Fourier coefficients). For every $n \geq 1$,

$$c_{4^n} = \frac{c_1}{4^n}, \quad \text{so} \quad \limsup_{k \rightarrow \infty} k |c_k| = |c_1| > 0.$$

Proof. Take $\alpha = 4^n$ in the recursion. Then $A(4^n) = A(0)$ (the exponentials $e^{-2\pi i 4^{n-1} j}$ are all 1), $J(4^{n-1}) = 0$ (a full number of oscillations), and $B(4^n) = B(0) = \frac{1}{3}(1 + \omega + \bar{\omega} + 1) = \frac{1}{3}(2 + 2 \cos \frac{\pi}{3}) = 1$. Hence $F(4^n) = \frac{1}{4} B(0) F(4^{n-1}) = \frac{1}{4} F(4^{n-1})$, and iterating, $c_{4^n} = F(4^n) = 4^{-n} F(1) = c_1/4^n$. ■

This is the key spectral fact: the self-similarity (expansion $\times 4$, contraction $\times \frac{1}{3}$) **forces** a Hadamard-gap subsequence of Fourier coefficients, with $|c_{4^n}| = |c_1|/4^n$ decaying exactly like k^{-1} along that subsequence. The subsequence $\{4^n\}$ is a **Hadamard gap** sequence ($4^{n+1}/4^n = 4 \geq 1 + \eta$), and the identity places the Koch parametrization squarely in the theory of **lacunary (Hadamard-gap) series** — the same theory that underlies the Weierstrass function and the classical results on natural boundaries of lacunary analytic functions.

The script verifies the identity to machine precision and the generic decay:

```

symmetry  phi(1-t) = 1 - conj(phi(t)):  max error = 1.31e-13
c_1 = -0.000000 + 0.083968 j      |c_1| = 0.083968 (purely imaginary)
lacunary identity  c_{4^n} = c_1 / 4^n (exact, from the functional equation):
  n= 1  k=      4   |c_k|*k = 0.083968   c_k/(c_1/4^n) = 1.00000000
  n= 2  k=     16   |c_k|*k = 0.083968   c_k/(c_1/4^n) = 1.00000000
  ...
  n=12  k= 16777216 |c_k|*k = 0.083968   c_k/(c_1/4^n) = 1.00000000
generic decay, log-log fit over k in [16, 4096]: -0.9959 (envelope O(k^-1))
cross-check vs FFT (N = 2^15, expect O(1/N) relative error):
  k=   1: exact |c_k| = 8.396827e-02   FFT |c_k| = 8.396966e-02   rel. diff =
    ↪ 0.00%
  k=  16: exact |c_k| = 5.248017e-03   FFT |c_k| = 5.249414e-03   rel. diff =
    ↪ 0.03%
  k= 256: exact |c_k| = 3.280011e-04   FFT |c_k| = 3.295453e-04   rel. diff =
    ↪ 0.47%
  k=4096: exact |c_k| = 2.050007e-05   FFT |c_k| = 2.420459e-05   rel. diff =
    ↪ 18.07%
```

Two further observations, both verified:

- **Symmetry.** The curve is symmetric about the vertical line $x = \frac{1}{2}$: $\varphi(1-t) = 1 - \overline{\varphi(t)}$ (error 1.3×10^{-13}). This forces every c_k to be *purely imaginary* (the real part vanishes), so $c_1 = 0.083968 i$.
- **Not absolutely convergent.** The envelope is $O(k^{-1})$ (fit slope -0.996), and the absolute sum grows like $C \log N$:

```

absolute convergence test (sum |c_k| over k=1..N):
  N = 100: sum |c_k| = 0.736809
  N = 400: sum |c_k| = 0.956341
  N = 1600: sum |c_k| = 1.176666
```

$N = 6400: \quad \text{sum } c_k = 1.397212$
--

so $\sum_k |c_k| = \infty$ — the Fourier series of φ is *not* absolutely convergent, as expected for a function whose coefficients decay only as $O(k^{-1})$. (The lacunary subsequence $\sum_n |c_{4^n}| = |c_1| \sum_n 4^{-n}$ *does* converge; the divergence comes from the bulk of the coefficients.)

Remark (regularity vs. decay). A generic $C^{0,\alpha}$ function has coefficients $O(k^{-\alpha})$; here $\alpha = \log 3 / \log 4 \approx 0.7925$ but the observed envelope is the *faster* $O(k^{-1})$. The self-similarity gives better-than-generic decay, yet it is still *sharp* (the lacunary part $c_1/4^n$ prevents any $o(k^{-1})$ decay). This interplay between the Hölder exponent and the Fourier envelope is a recurring theme in the study of self-similar and rough functions.

8. The Koch snowflake: perimeter, area, and the quasicircle

8.1 Perimeter and area

Apply the construction (with *outward* bumps) to the three sides of an equilateral triangle of side 1. Let P_n and A_n be the perimeter and enclosed area at level n .

- **Perimeter.** $P_n = 3(4/3)^n \rightarrow \infty$ — the same infinite-length phenomenon as §6.2.
- **Area.** At step $n \geq 1$ one adds $3 \cdot 4^{n-1}$ triangles of side 3^{-n} , each of area $(\sqrt{3}/4)9^{-n}$. Hence

$$A_\infty = \frac{\sqrt{3}}{4} \left[1 + \sum_{n=1}^{\infty} 3 \cdot 4^{n-1} 9^{-n} \right] = \frac{\sqrt{3}}{4} \left[1 + \frac{3}{4} \sum_{n=1}^{\infty} \left(\frac{4}{9} \right)^n \right] = \frac{\sqrt{3}}{4} \left(1 + \frac{3}{5} \right) = \frac{8}{5} \cdot \frac{\sqrt{3}}{4} = \frac{2\sqrt{3}}{5}.$$

So the snowflake has **finite area** $A_\infty = \frac{8}{5}A_0 \approx 0.6928$ but **infinite perimeter** — the classical “infinite coastline, finite land” paradox. The script confirms the convergence:

Koch snowflake (equilateral, side 1): $A_0 = \text{sqrt}(3)/4 = 0.433013$				
n	perimeter P_n	$3 \cdot (4/3)^n$	area A_n	A_n/A_0
0	3.000000	3.000000	0.433013	1.00000
1	4.000000	4.000000	0.577350	1.33333
2	5.333333	5.333333	0.641500	1.48148
...				
10	53.273180	53.273180	0.692742	1.59982
limit:	$A = 2 \cdot \text{sqrt}(3)/5 = 0.692820 = (8/5) \cdot A_0 \quad (\text{ratio } 1.60000)$			

Koch snowflake

Figure 2. The Koch snowflake at increasing levels: $\text{perimeter} \rightarrow \infty$, $\text{area} \rightarrow \frac{8}{5}A_0$.

8.2 The snowflake as a quasicircle

For the expert reader, the closed Koch curve is a canonical example in geometric function theory. A **quasicircle** is the image of the unit circle under a quasiconformal map of the plane; equivalently (Näkki-Väisälä), a Jordan curve is a quasicircle iff it has **bounded turning**, and the parametrization can be chosen **quasisymmetric**. The bi-Hölder estimate of §6.1 — $c|t - s|^\alpha \leq |\varphi(t) - \varphi(s)| \leq C|t - s|^\alpha$ — is exactly a quasisymmetry-type condition, and it places the Koch curve among the quasicircles. It is a standard fact (and follows from the bounded-turning criterion) that the Koch snowflake is a quasicircle, and indeed a quasicircle that is *not* bilipschitz to the circle (its dimension exceeds 1). See e.g. [Väisälä, 1978] and the survey literature on quasicircles.

9. Numerical approximation and convergence (numerical analysis)

Two questions of practical interest close the essay.

9.1 Rate of the canonical iteration

As shown in §3.1, the sup-norm error of the n -th polygonal approximation is *exactly* $E_n = 3^{-n}E_0$ with $E_0 = \sqrt{3}/6$. So the naive iteration is a **linear** (geometric, ratio $1/3$) method: each extra level buys a factor of 3 in accuracy at a factor of 4 in the number of segments.

9.2 Approximation complexity

A more informative quantity is the *cost* of a target accuracy ε . Because the curve has box dimension $d = \log 4 / \log 3$, an ε -net needs $\sim \varepsilon^{-d}$ points, and the number of segments of P_n needed to reach $\varepsilon = 3^{-n}$ is $4^n = (3^n)^{\log 4 / \log 3} = \varepsilon^{-d}$. Thus the cost grows like $\varepsilon^{-\log 4 / \log 3} \approx \varepsilon^{-1.262}$, strictly worse than the ε^{-1} of a C^1 curve. This is the numerical-analytic signature of the fractal: the “roughness” (dimension > 1) is precisely what degrades the approximation rate.

```
figures written to ../figures/
fig1_iterations.png      (the polygonal approximations P_n)
fig2_snowflake.png       (the snowflake, perimeter/area)
fig3_dimension_holder.png (box-counting dimension + Holder exponent)
fig4_fourier_pvar.png    (Fourier decay + p-variation threshold)
fig5_convergence_complexity.png (convergence rate + approximation cost)
```

Dimension and Holder exponent

Figure 3. Left: box-counting dimension ($\log\text{-}\log$ slope $\rightarrow \log 4 / \log 3$). Right: the exact Hölder exponent $\log 3 / \log 4$ measured from probe pairs.

Fourier decay and p-variation

Figure 4. Left: Fourier decay of φ (envelope $O(k^{-1})$, lacunary peaks at $k = 4^n$). Right: the p -variation threshold $p^* = \log 4 / \log 3$.

Convergence and complexity

Figure 5. Left: the exact geometric convergence $E_n = 3^{-n}E_0$. Right: the fractal approximation complexity $\varepsilon^{-\log 4 / \log 3}$ versus the smooth ε^{-1} .

10. Summary of the principal results

Quantity	Value	Theorem / source
Existence & convergence	$P_n \rightarrow \varphi$ uniformly, $\ \varphi - P_n\ _\infty = 3^{-n}E_0$	Banach fixed point / Weierstrass M-test (§3)
Tangent	none at any point	Koch (1904) (§4)
Hausdorff dimension	$\log 4 / \log 3 \approx 1.2619$	Hutchinson-Falconer, OSC (§5)
Box-counting dimension	$\log 4 / \log 3$ (content ≈ 1.02)	§5
Hölder regularity	$C^{0,\alpha}$, $\alpha = \log 3 / \log 4 \approx 0.7925$ (sharp)	§6.1
Arc length	∞	§6.2

p -variation	finite iff $p \geq \log 4 / \log 3$ (critical value $= 1$, attained by vertex partitions)	§6.3
Fourier (lacunary)	$c_{4^n} = c_1/4^n, c_1 = 0.083968 i, envelope O(k^{-1}), \sum c_k = \infty$	§7
Snowflake perimeter / area	$P_n = 3(4/3)^n \rightarrow \infty / A_\infty = \frac{8}{5} A_0$	§8
Quasicircle	yes (bounded turning, quasisymmetric)	§8.2

The unifying theme is **self-similarity**. A single structural fact — the curve is four exact copies of itself at ratio $1/3$ — simultaneously determines the dimension ($\log 4 / \log 3$), the Hölder exponent ($\log 3 / \log 4$, its reciprocal), the p -variation threshold (again $\log 4 / \log 3$), and the lacunary Fourier spectrum ($c_{4^n} = c_1/4^n$). The Koch curve is thus the cleanest possible laboratory for the interplay between geometric self-similarity and function-analytic regularity.

References

1. **H. von Koch**, *Sur une courbe continue sans tangente, obtenue par une construction géométrique élémentaire*, Arkiv för matematik, astronomi och fysik **1** (1904), 681–704. (The original paper.)
2. **J. E. Hutchinson**, *Fractals and Self-Similarity*, Indiana Univ. Math. J. **30** (1981), 713–747. (The self-similar-set / fixed-point theorem used in §3.2 and §5.)
3. **K. Falconer**, *Fractal Geometry: Mathematical Foundations and Applications*, Wiley, 1990. (Hausdorff and box dimensions, the open set condition; standard reference.)
4. **P. Mattila**, *Geometry of Sets and Measures in Euclidean Spaces*, Cambridge Univ. Press, 1995. (Hausdorff measure and dimension.)
5. **S. J. Taylor**, *Fractal Sets*, Cambridge Univ. Press, 1995. (Dimensions of curves, p -variation, rectifiability.)
6. **J.-P. Kahane**, *Some Random Series of Functions*, 3rd ed., Cambridge Univ. Press, 1985. (Lacunary / Hadamard-gap series, §7.)
7. **J. Väisälä**, *Lectures on n -Dimensional Quasiconformal Mappings*, Springer, 1978. (Quasicircles, bounded turning, quasisymmetry, §8.2.)
8. **T. J. Lyons, M. Caruana, T. Lévy**, *Differential Equations Driven by Rough Paths*, in *École d'été de probabilités de Saint-Flour XXXIV—2004* (ed. J. Picard), Lecture Notes in Mathematics **1908**, Springer, 2007, 1–119. (The p -variation / rough-paths viewpoint, §6.3.)

All numerical tables and figures in this essay were generated by `koch_simulation.py` (Python 3, NumPy, Matplotlib). The exact identities — $\varphi(4^{-n}) = 3^{-n}$, $c_{4^n} = c_1/4^n$, $\|\varphi - P_n\|_\infty = 3^{-n} E_0$, the symmetry $\varphi(1-t) = 1 - \overline{\varphi(t)}$ — are verified to machine precision; the statistical estimates (box counts, Hölder exponents, Fourier envelope) are verified against their closed-form limits.